		Spill		proportion spoiled	
1.	A - $\frac{2}{5}$		$\frac{1}{20}$		$\frac{1}{50}$
	B - $\frac{3}{10}$		$\frac{1}{25}$		$\frac{3}{250}$
	C - $\frac{1}{5}$		$\frac{1}{50}$		$\frac{1}{250}$
	D - $\frac{1}{10}$		$\frac{1}{100}$		$\frac{1}{1000}$

Σ 3.7%

(a) 3.7 % (see above right)

(b) 3.7 % spoiled

2 % of spoiled products were from A

$20/37 \approx 0.541$

2. (a) (i) $\frac{11+4}{11+4+37+139} = \frac{15}{191} \approx 7.9\%$

(ii) $\frac{53}{53+414+16} = \frac{53}{483} \approx 11.0\%$

(b) (i) $\frac{11}{11+37} = \frac{11}{48} \approx 22.9\%$

(ii) 0%

(c) $\frac{11}{53+11+4} = \frac{11}{68} = 16.2\%$

(d) among all 674 murders represented, 191 are represented w/ black defendants, meaning that the proportion of these defendants represented is 28.3%

3. $P(A) = 0.2$ $\qquad$ $P(B \cap C) = P(B) \times P(C)$

$P(B) = 0.4$

$P(A \cup B \cup C) = 0.9$

$A \cap B = \emptyset$

$P(A \cap C) = P(A) \times P(C)$

Inclusion - exclusion formula

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap C) - P(B \cap C)$$
$$0.9 = 0.2 + 0.4 + P(C) - 0.2P(C) - 0.4P(C)$$
$$- \underset{0}{P(A \cap B)} + \underset{0}{P(A \cap B \cap C)}$$

$$0.9 = 0.6 + 0.4P(C)$$
$$9 = 6 + 4P(C)$$
$$3 = 4P(C)$$
$$P(C) = 3/4 = 0.75 = 75\%$$

4. P stands for proportion below:

$P(d) = 0.36$

$P(c|d) = 0.22$ $\qquad$ (a) 7.92%

$P(c) = 0.3$

$P(d \cap c) = P(d) \times P(c|d) = 0.0792$

$P(d|c) = P(d \cap c) / P(c) = 0.264$

(b) 26.4%